Probabilistic Thinking

A Primer

Edward Milsom

2026-09-24

Slide 1 **In this Episode ...**

- A little motivation
- Definitions of probability
  - discrete and continuous
  - univariate and multivariate
- The two rules ... and Bayes
- Concepts in multivariate probability
- Probabilistic modelling and machine learning
- Some useful probability distributions

Introduction & Motivation

Slide 2 **Theory**

- Laplace's *Théorie Analytique des Probabilités* (Analytic Theory of Probability), was first published in 1812
  - his *Essai Philosophique sur les Probabilités* (1814) from Lecture 1 was intended as an "accessible" summary for the general public
- Depending on the edition, the *Théorie Analytique* can total 846 pages
- We'll just cover the main bits ...

Slide 3 **Alternative Reading**

- A more formal 50-page introduction: **Chapter 6** of (Marc Peter Deisenroth and Ong 2020)
- For a more succinct primer, see: **Appendix C** of (Prince 2023)

- Also, **Section 1.2** of: (Bishop 2006)

Slide 4 **Motivation: January 28, 1986 ...**



Slide 5 **“Nature Cannot Be Fooled” (Richard Feynman)**

“A logarithmic plot suggested a straight line, so it was supposed that the erosion varied as the .58 power of the heat, the .58 being determined by a nearest fit. At any rate, adjusting some other numbers, it was determined that the model agreed with the erosion (to depth of one-third the radius of the ring).
There is nothing much so wrong with this as believing the answer!
Uncertainties appear everywhere …
When using a mathematical model, careful attention must be given to uncertainties in the model.” *Professor Richard Feynman — excerpt from Appendix F of the Report of the*
Presidential Commission on the Space Shuttle Challenger Accident (1986).

Slide 6 **It Isn't Rocket Science ...**

Thomas Bayes' original question, published in 1763:

> PROBLEM.
>
> *Given* the number of times in which an unknown event has happened and failed: *Required* the chance that the probability of its happening in a single trial lies somewhere between any two degrees of probability that can be named.

Probability

Slide 7 **Discrete Probability**

- A discrete random variable x takes on a fixed set of values. *e.g.*
  - $\{0, 1\}$ — perhaps the output of a binary classifier
  - $\{A_1, A_2, A_3, A_4, A_5\}$ — perhaps some categorical input feature
- We can define the probability $P\left(x = A_j\right) = \theta_j$, with:

$$0 \leq \theta_j \leq 1$$
$$\sum_j \theta_j = 1$$

- The quantity $P(x)$ is referred to as the probability mass function
- Informally, an upper-case $P(\cdot)$ is often adopted to denote probability *mass*

Slide 8 **Continuous Probability**

- A continuous random variable x is typically defined over (a subset of) the real line
- It is characterised by its probability density function (p.d.f.) $p(x)$, where:

$$p(x) \geq 0$$
$$\int_{-\infty}^{+\infty} p(x)\, dx = 1$$

  from which a desired probability mass can be calculated:

$$P(a \leq x \leq b) = \int_a^b p(x)\, dx$$

- Note that $P(x = a) = 0$ and $p(x)$ can be greater than one!

- Informally, a lower-case $p(\cdot)$ is often adopted to denote probability *density*

Slide 9 **Joint Probability: Multiple Variables**

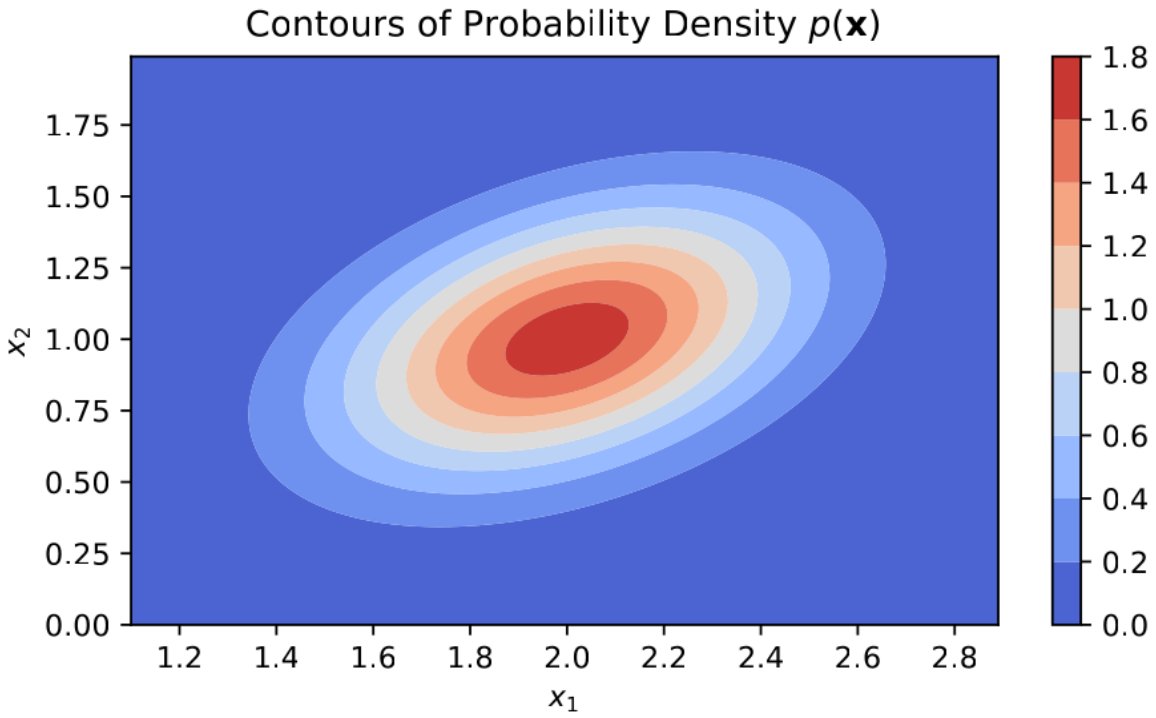
- In general, the interesting scenarios are where we wish to manipulate (and infer!) *multiple* random variables: *e.g.* x_1 and x_2 in the simplest case
- The joint distribution (joint p.d.f.) characterises the propensity for multiple variables to take on particular combinations of values: $p(x_1, x_2)$
- For succinctness, when handling many joint variables, we adopt vector notation:
  - $\mathbf{x} \equiv (x_1, x_2)$ and so $p(x_1, x_2) \rightarrow p(\mathbf{x})$
- In this case:

$$\begin{aligned} \int \int p(x_1, x_2)\, dx_1 dx_2 &= 1 \\ \text{or } \int p(\mathbf{x})\, d\mathbf{x} &= 1 \end{aligned}$$

- Note: $\mathbf{x}$ may contain a mix of continuous and discrete variables

---

Slide 10 **An Example Joint Continuous Distribution**



Slide 11 **The Two Rules of Probability**

The sum rule (marginalisation or integrating out):

- If variable x_2 is an element of some discrete set $\mathcal{X}$ then:

$$P(x_1) = \sum_{x_2 \in \mathcal{X}} P(x_1, x_2)$$

- If x_2 is continuous in $\mathcal{R}$ then:

$$p(x_1) = \int_{\mathcal{R}} p(x_1, x_2)\, dx_2$$

The product rule:

$$\begin{aligned} p(x_1, x_2) &= p(x_1|x_2)\, p(x_2), \text{ or equivalently} \\ &= p(x_2|x_1)\, p(x_1) \end{aligned}$$

As a consequence of the product rule we get Bayes' rule ...

Slide 12 **Bayes' Rule: A Corollary of The Two Rules of Probability**

Bayes' Rule

$$p(x_2|x_1) = \frac{p(x_1|x_2)\, p(x_2)}{p(x_1)}$$

In the context of machine learning models, this will often be applied and interpreted as:

$$p(\text{Unknown}|\text{Data}) = \frac{p(\text{Data}|\text{Unknown})\, p(\text{Unknown})}{p(\text{Data})}$$

Slide 13 **Key Concepts in Joint Probability**

As well as the joint distribution, we are also interested in two further quantities.

Marginal Distribution

- If we "ignore" x_2, then the distribution of x_1 alone is the marginal distribution $p(x_1)$
  - it is obtained from the joint by applying the sum rule

Conditional Distribution

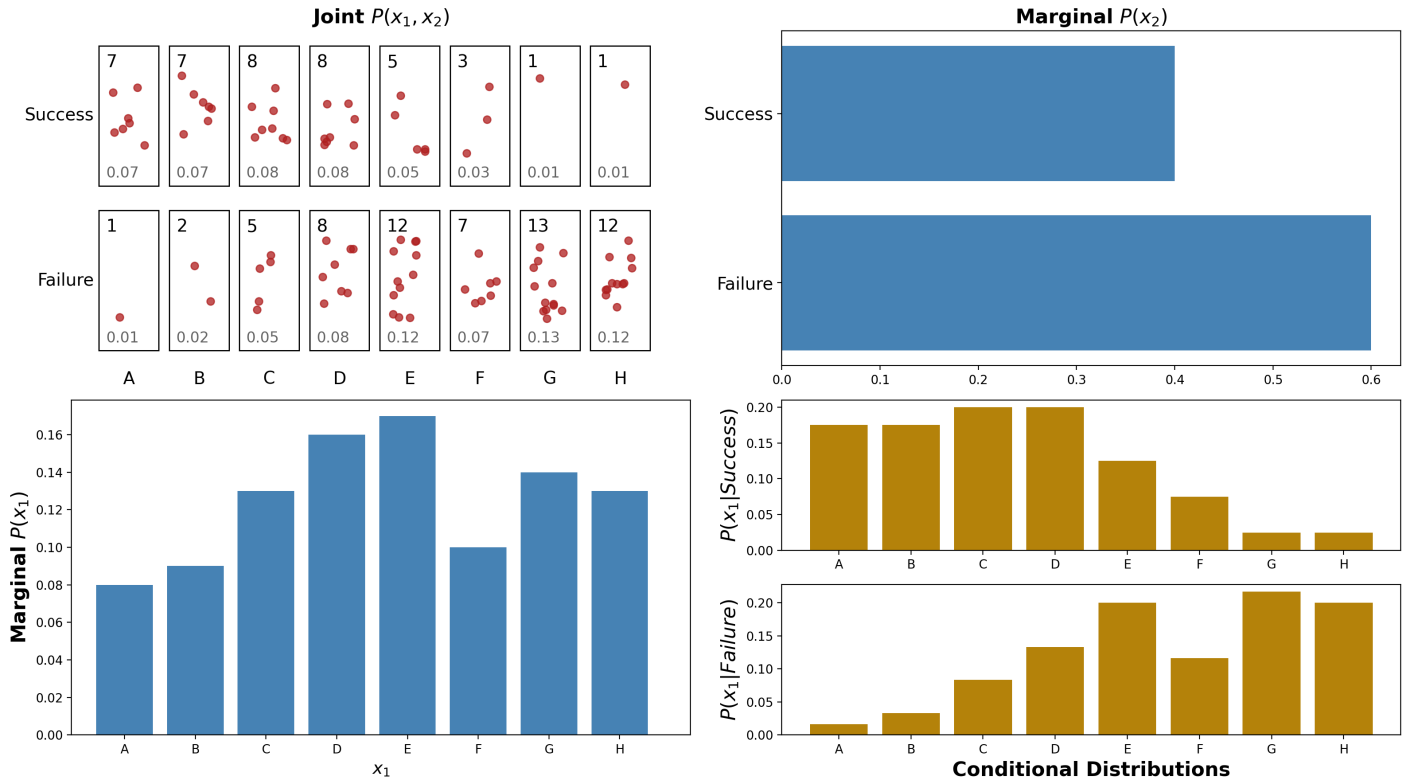
- The distribution of x_1 values given a fixed value of x_2 is the conditional distribution $p(x_1|x_2)$
  - it is effectively a normalised "slice" across the joint distribution

Also ... **Independence**

- If x_1 does not depend on x_2, then $p(x_1|x_2) = p(x_1)$ and $p(x_1, x_2) = p(x_1)\, p(x_2)$

The next four slides illustrate these concepts ...

Slide 14 Marginals & Conditionals for Discrete Distributions



Slide 15 **Marginals (“Understanding Deep Learning”, Appendix C)**

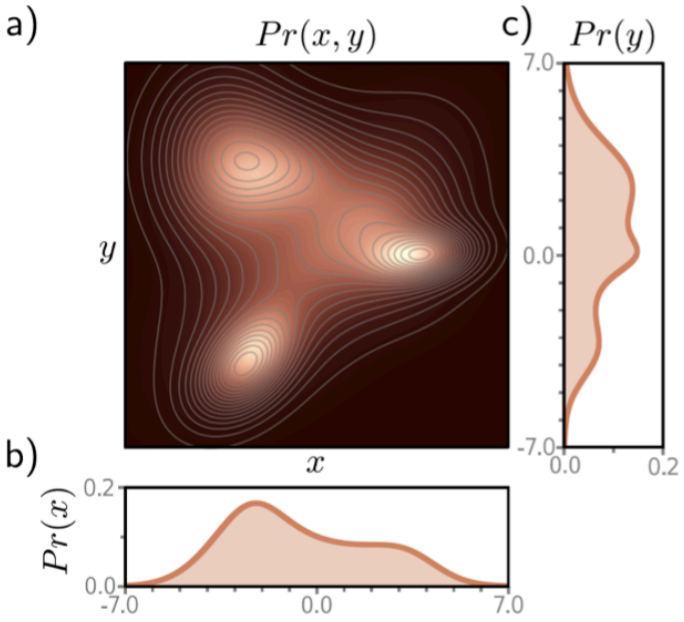


Figure C.1 Joint and marginal distributions. a) The joint distribution $Pr(x, y)$ captures the propensity of variables x and y to take different combinations of values. Here, the probability density is represented by the color map, so brighter positions are more probable. For example, the combination $x = 6, y = 6$ is much less likely to be observed than the combination $x = 5, y = 0$. b) The marginal distribution $Pr(x)$ of variable x can be recovered by integrating over y. c) The marginal distribution $Pr(y)$ of variable y can be recovered by integrating over x.

This work is subject to a Creative Commons CC-BY-NC-ND license. (C) MIT Press.

Slide 16 **Conditionals (“Understanding Deep Learning”, Appendix C)**

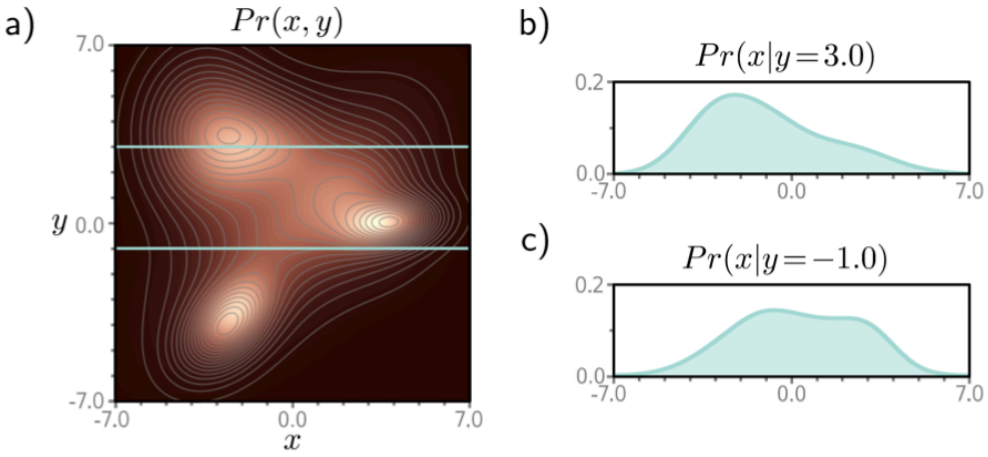


Figure C.2 Conditional distributions. a) Joint distribution $Pr(x, y)$ of variables x and y. b) The conditional probability $Pr(x|y = 3.0)$ of variable x, given that y takes the value 3.0, is found by taking the horizontal “slice” $Pr(x, y = 3.0)$ of the joint probability (top cyan line in panel a), and dividing this by the total area $Pr(y = 3.0)$ in that slice so that it forms a valid probability distribution that integrates to one. c) The joint probability $Pr(x, y=-1.0)$ is found similarly using the slice at $y=-1.0$.

This work is subject to a Creative Commons CC-BY-NC-ND license. (C) MIT Press.

Slide 17 **Independence ("Understanding Deep Learning", Appendix C)**

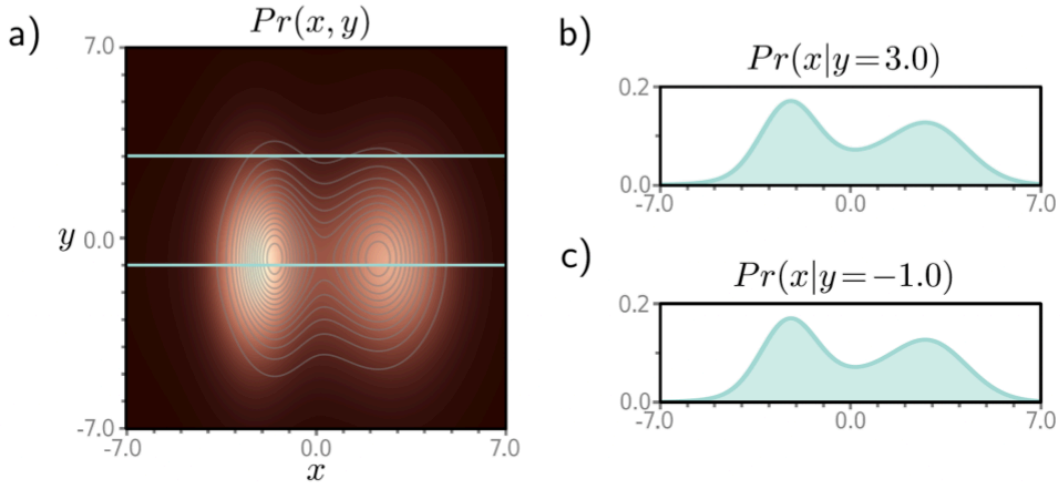


Figure C.3 Independence. a) When two variables x and y are independent, the joint distribution factors into the product of marginal distributions, so $Pr(x, y) = Pr(x)Pr(y)$. Independence implies that knowing the value of one variable tells us nothing about the other. b–c) Accordingly, all of the conditional distributions $Pr(x|y = \bullet)$ are the same and are equal to the marginal distribution $Pr(x)$.

This work is subject to a Creative Commons CC-BY-NC-ND license. (C) MIT Press.

Data Sets, Probability & Likelihood

Slide 18 **Data Sets and Probabilistic Modelling**

- When adopting a probabilistic modelling (not necessarily Bayesian) approach to machine learning, we often need to compute the probability of the data under some model which has parameters $\boldsymbol{\theta} = \left(\theta_1, \theta_2, ..., \theta_M\right)$
- Perhaps we have a data set comprising N example vectors: $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, ..., \mathbf{x}_N\}$
- The probability of the entire set, under the model given the parameters, is the conditional joint distribution $p\left(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, ..., \mathbf{x}_N|\boldsymbol{\theta}\right)$
- If we assume (we often do!) that the data samples are all independent and identically distributed, or i.i.d., then this simplifies to:

$$p\left(\text{Data}|\boldsymbol{\theta}\right) = \prod_{n=1}^{N} p\left(\mathbf{x}_n|\boldsymbol{\theta}\right)$$

Slide 19 **Data Sets and Likelihood**

- As just shown, the formula $p\left(\text{Data}|\boldsymbol{\theta}\right)$ is generally interpreted as the probability of the data when viewed as **a function of all $\mathbf{x}_n$ with the parameters $\boldsymbol{\theta}$ fixed**

- However, *the exact same expression* is referred to as the likelihood of the parameters when viewed as **a function of all $\boldsymbol{\theta}$ with the data $\mathbf{x}_n$ fixed**

- It is **important** to be aware that:

$$\begin{aligned} \int \cdots \int \prod_{n=1}^{N} p\left(\mathbf{x}_n|\boldsymbol{\theta}\right) d\mathbf{x}_1 \ldots d\mathbf{x}_N &= 1, \\ \text{but } \int \prod_{n=1}^{N} p\left(\mathbf{x}_n|\boldsymbol{\theta}\right) d\boldsymbol{\theta} &\neq 1 \end{aligned}$$

Slide 20 **Machine Learning, Maximum Likelihood & Logarithms**

- When fitting models (both in statistics and machine learning), we often apply the principle of maximum likelihood:
  - that is, maximising $p\left(\text{Data}|\boldsymbol{\theta}\right) = \prod_{n=1}^{N} p\left(\mathbf{x}_n|\boldsymbol{\theta}\right)$ with respect to $\boldsymbol{\theta}$

- When modelling data, and particularly for maximising likelihood, it is often more mathematically convenient to work with the log-likelihood:

$$\log p\left(\text{Data}|\boldsymbol{\theta}\right) = \sum_{n=1}^{N} \log p\left(\mathbf{x}_n|\boldsymbol{\theta}\right)$$

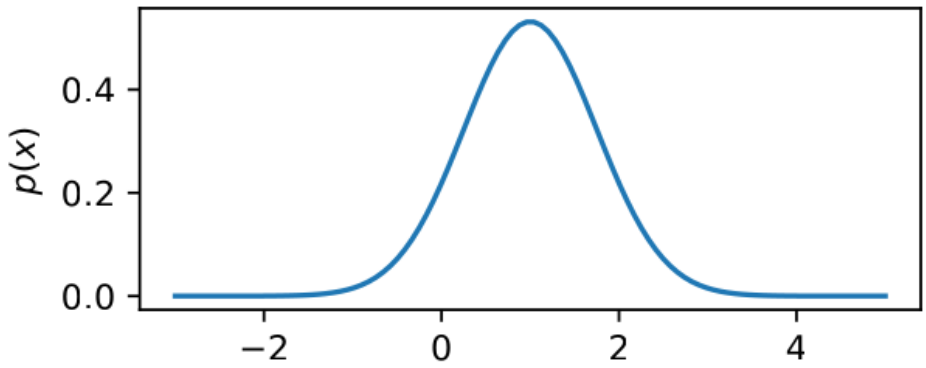
- A key point to note is that the logarithm function is monotonic (increasing)
  - "monotonic increasing" implies that $\log\left(z_2\right) > \log\left(z_1\right)$ if $z_2 > z_1$
  - and so the value of $\boldsymbol{\theta}$ that maximises $\log p\left(\text{Data}|\boldsymbol{\theta}\right)$ is therefore identical to that which maximises $p\left(\text{Data}|\boldsymbol{\theta}\right)$

Examples of Some Common Distributions

Slide 21 **Gaussian, or Normal, Distribution: Continuous in $[-\infty, +\infty]$**

$$p\left(x|\mu,\sigma^2\right) = \left(2\pi\sigma^2\right)^{-1/2} \exp\left\{-\frac{\left(x-\mu\right)^2}{2\sigma^2}\right\}$$

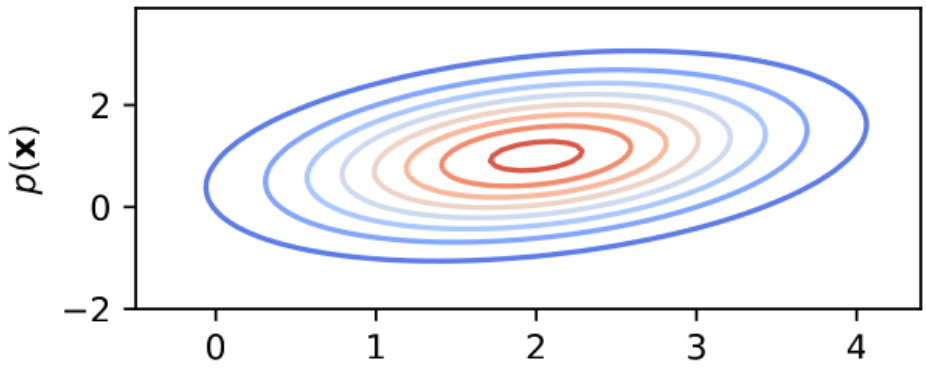
```
gaussian = scipy.stats.norm(loc=1.0, scale=0.75)
x = np.linspace(start=-3, stop=5, num=100)
plt.figure(figsize=(4, 1.75))
plt.plot(x, gaussian.pdf(x))
plt.ylabel('$p(x)$')
```



Slide 22 **The Multivariate Gaussian (Normal)**

$$p(x\left|\boldsymbol{\mu},\boldsymbol{\Sigma}\right.) = (2\pi)^{-D/2}\left|\boldsymbol{\Sigma}\right|^{-1/2}\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

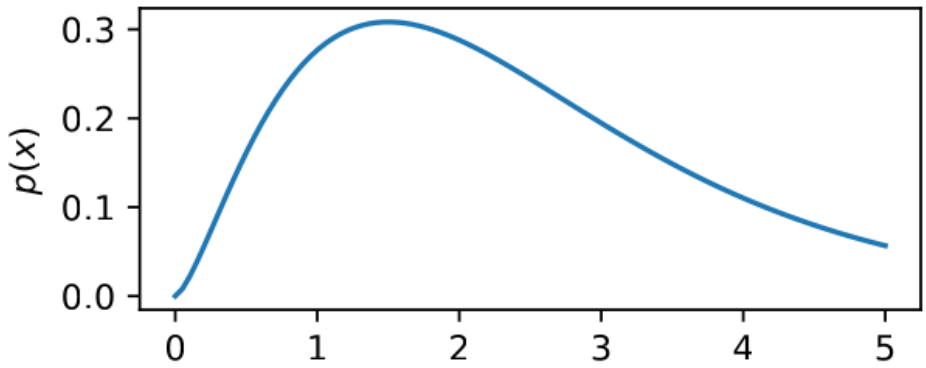
```
mg = scipy.stats.multivariate_normal(mean=[2, 1], cov=[[1,0.3],[0.3,1]])
x, y = np.mgrid[-0.5:4.5:0.1, -2:4:0.1]
plt.figure(figsize=(4, 1.75))
plt.contour(x, y, mg.pdf(np.dstack((x, y))), levels=8, cmap="coolwarm")
plt.ylabel('$p(\\mathbf{x})$')
```



Slide 23 **Gamma Distribution: Continuous in** $[0, +\infty]$

$$p\left(x|a,\lambda\right) = \frac{\lambda^{a}}{\Gamma(a)}x^{a-1}e^{-\lambda x}$$

```
gamma = scipy.stats.gamma(a=2.5, scale=1)
x = np.linspace(start=0, stop=5, num=100)
plt.figure(figsize=(4, 1.75))
plt.plot(x, gamma.pdf(x))
plt.ylabel('$p(x)$')
```



Slide 24 **Bernoulli Distribution: Discrete in** $\{0, 1\}$

- The Bernoulli is used to describe binary variables, *e.g.* classifier outputs:

$$P\left(x|\theta\right) = \begin{cases} \theta & \text{if } x = 1 \\ 1 - \theta & \text{if } x = 0 \end{cases}$$

- This can be more concisely written as:

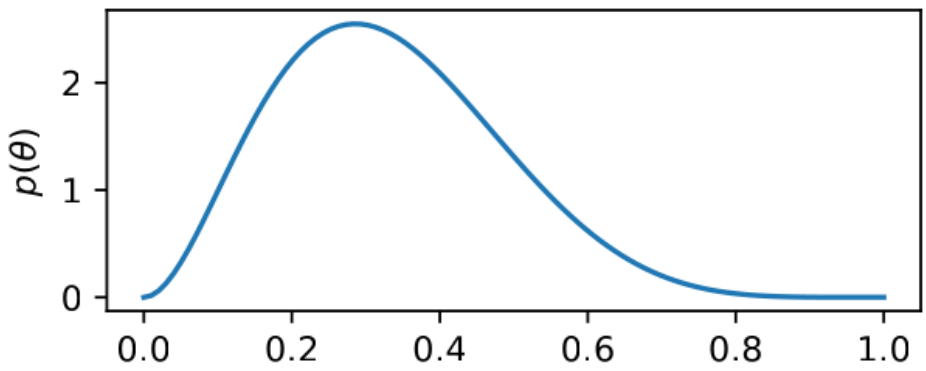
$$P\left(x|\theta\right) = \theta^{x}\ \left(1 - \theta\right)^{1 - x}$$

- The generalisation of the Bernoulli distribution to a variable with more than two outcomes is the categorical distribution, where $P\left(x = A_j\right) \equiv \theta_j$

Slide 25 **Beta Distribution: Continuous in** $[0, 1]$

$$p\left(\theta|a, b\right) = \frac{\Gamma\left(a + b\right)}{\Gamma\left(a\right)\ \Gamma\left(b\right)}\theta^{a - 1}\left(1 - \theta\right)^{b - 1}$$

```
beta = scipy.stats.beta(a=3, b=6)
x = np.linspace(start=0, stop=1, num=100)
plt.figure(figsize=(4, 1.75))
plt.plot(x, beta.pdf(x))
plt.ylabel(r'$p(\theta)$')
```



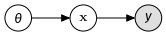
If Time Allows ...

Slide 26 Looking Ahead ... Conjugacy of Distributions

- The Beta distribution will prove useful in Bayesian analysis
- It is the conjugate prior distribution for the θ parameter in the Bernoulli distribution
- What does this mean?
- Find out in next Tuesday's lecture ...

Slide 27 Looking Further Ahead ... Probabilistic Graphical Models

- When thinking probabilistically, it can be helpful to visualise our models using probabilistic graphical model (PGM) format



- This specifies a joint distribution that factorises thus:

$$p\left(y, \mathbf{x}, \theta\right) = p\left(y|\mathbf{x}\right)\, p\left(\mathbf{x}|\theta\right)\, p\left(\theta\right)$$

  - shaded nodes (y) are observed
  - clear nodes ($\mathbf{x}$, θ) are unknowns, or latent variables
  - arrows indicate conditional dependencies

- We'll return to this in more detail in the context of hierarchical models

---

Slide 28 Underlying Principles of Dependence Between Variables

Independence	Observed x and y are independent: $p\left(x, y\right) = p\left(x\right)\, p\left(y\right)$ **Direct Dependence**	Let's say we know y depends on x directly:	Observed x and y are dependent: $p\left(x, y\right) = p\left(y|x\right)\, p\left(x\right)$	**Indirect Dependence**	Observe ally inde $p\left(x, y,$

Bishop, Christopher M. 2006. *Pattern Recognition and Machine Learning*. Springer. https://www.microsoft.com/en-us/research/people/cmbishop/#!prml-book.
Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong. 2020. *Mathematics for Machine Learning*. Cambridge University Press. https://mml-book.com/.
Prince, Simon J. D. 2023. *Understanding Deep Learning*. MIT Press. http://udlbook.com.